AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956
COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I041*

Team Name: *Aura*

Domain: *Domain 6: Healthcare, Biotechnology & Human Well-being*

Amrita Vishwa Vidyapeetham, Coimbatore Campus

Date: 11th October 2026

Presentation Outline

1. Problem Statement
2. Problem Understanding
3. Existing Gaps & Challenges
4. Our Proposed Solution
5. Technical Approach
6. Innovation & Uniqueness
7. Feasibility & Scalability
8. Use Cases
9. Impacts and Benefits
10. Sustainable Development Goals
11. Individual Member Contribution

Problem Statement

Genetic variation influences a person's susceptibility to disease. But turning raw DNA sequence data into understandable, evidence-based risk information is still a complex computational and medical challenge.

How can we build an AI-assisted pipeline that analyses DNA sequence data, identifies and annotates relevant variants, estimates disease risk where the evidence supports it, and reports the findings together with their evidence, uncertainty, and limitations?

Problem Understanding

- **Who:** genomics researchers, genetic counsellors, and clinicians who need to make sense of sequencing results.
- **Cause:** risk for most common diseases comes from many variants with small effects, and the steps from raw reads to a readable result are spread across separate tools.
- **Effect:** findings get misread, risk is overstated or missed, and results are weakest for populations that are under-represented in genomic data.

Existing Gaps & Challenges

- **Implementation:** genomic processing, variant interpretation, risk prediction, and reporting rarely sit in one reproducible workflow.
- **Interpretability:** outputs are often not linked to the specific variants and published evidence behind them.
- **Population generalizability:** most risk models were built on European-ancestry data and may not hold for Indian populations.
- **Validation and reporting:** performance, calibration, and uncertainty are often not communicated in an understandable way.

Our Proposed Solution

- An AI-assisted genomic analysis system that turns DNA sequence data into an evidence-linked disease-risk report.
- **Pipeline:** DNA data → Quality check → Identify genetic changes → Analyze disease links → Predict disease risk → Generate health report.
- **Risk models:** polygenic risk scores (clumping and thresholding, LDpred2, PRS-CS), a logistic regression baseline, and neural networks only if the data and validation justify them.
- **Boundary:** research and informational support. It does not diagnose, prescribe, or replace genetic counselling.

Technical Approach



Quality control and alignment, variant calling and annotation, then PRS and baseline models feed an evidence-linked report with uncertainty.

Tech Stack

Frontend: React

Backend: FastAPI

Database: PostgreSQL (restricted access)

AI/ML: scikit-learn, PLINK, GATK

Innovation & Uniqueness

- **Integrated:** one reproducible workflow from sequence data to report, instead of separate tools stitched together by hand.
- **Evidence-linked:** each finding points to its source (ClinVar, GWAS Catalog, PGS Catalog) and states its uncertainty.
- **Population-aware:** uses Indian genomes to check how risk scores behave in Indian populations.
- **Honest by design:** a PRS is reported as a genetic score, not a probability, unless it has been calibrated.

Follows ACMG/AMP variant interpretation (Richards et al., 2015) and TRIPOD+AI reporting (Collins et al., 2024).

Feasibility & Scalability

- **Feasibility:** Data: GenomeIndia (about 10,000 Indian whole genomes via IBDC), 1000 Genomes, gnomAD, ClinVar, GWAS Catalog, PGS Catalog.
- **Scalability:** diseases are added only as suitable data and validated models exist; workflows can run on the cloud (Nextflow or Snakemake).
- **Risks:** GenomeIndia needs approved access and comes from healthy individuals, so it supports allele-frequency and score-distribution checks, not disease labels. Mitigation: start with open data and published PRS weights, apply for access in parallel.

Use Cases (compulsorily 3 use cases must be there)

- **Early disease-risk prediction:** a person or clinic uploads genome data and gets risk estimates for diseases like type 2 diabetes or coronary artery disease, with the supporting variants and evidence.
- **Prevention planning:** the report supports a doctor's advice on earlier screening and lifestyle changes for people with higher genetic risk, before symptoms appear.
- **Personalized drug response:** variants that affect how a person handles medicines (for example CYP2C19 and clopidogrel) are flagged for the doctor to consider. It informs the doctor and does not prescribe.

Use Cases (continued)

- **Carrier screening:** families check for variants linked to inherited conditions relevant in India, such as beta-thalassemia and sickle cell disease.
- **Counselling support:** a genetic counsellor reviews candidate variants, evidence, and uncertainty in an evidence-linked report to prepare for a session.

Impacts and Benefits

- **Users:** clearer, evidence-linked genomic information with uncertainty stated, and less manual effort stitching tools together.
- **Social:** shows how well existing risk scores work for Indian populations, supporting more inclusive genomics.
- **Measurable outcomes:** a pipeline that runs end to end on public data, and models assessed on independent data for AUROC, sensitivity, specificity, and calibration.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 3	Target 3.4	Good Health and Well-being	Risk information to support prevention
SDG 9	Target 9.5	Industry, Innovation and Infrastructure	Applies AI to scientific research

Individual Member Contribution

Member	Contribution or responsibility
R Rohith Sai	Problem statement, understanding, and gaps
B Dinesh Kumar	Feasibility, scalability, use cases, and impacts
V Rohith	Solution design and genomics research
V Harivansh Reddy	Technical approach, datasets, and ML algorithms

Yukti ID

[IR2026-1051031]